THE ZERO LOCUS OF A MAP INTO A PURE FINITE PRESENTATION MODULE IS A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a reduction to scheme bases and an affine chart on which the module is projective, a constructibility and purity argument shrinking the base, and an application of the universally separating property

Editorial note. The label lemma-F-zero-closed-pure names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Universal purity of a flat module of finite presentation that is pure relative to the base, and the purity-on-top criterion relating purity to the associated points of the fibres
- The criterion reducing representability to representability over schemes, and etale locality of closed immersions and of descent data for sheaves
- The existence of a chart over an affine etale neighbourhood on which the pullback of the module is projective and whose image contains the fibre, the constructibility of the relative assassin, and the criterion that a map is injective exactly when its pullback along a universally separating morphism is
- The affine case for a projective module over the coordinate ring of an affine base

Source reference. Original lemma 03I0: sources/stacks-chow/tags.txt, source lines 4586–4586. Prerequisite pointer invoked at proof line(s) 1733; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02W5: sources/stacks-chow/tags.txt, source lines 3792–3792. Prerequisite pointer invoked at proof line(s) 1734; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 083L: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1627–1627. Prerequisite pointer invoked at proof line(s) 1779; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-flat.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 05KW: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 956–956. Prerequisite pointer invoked at proof line(s) 1745; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05L6: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1146–1146. Prerequisite pointer invoked at proof line(s) 1720; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 083J: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1335–1335. Prerequisite pointer invoked at proof line(s) 1784; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05MJ: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1269–1269. Prerequisite pointer invoked at proof line(s) 1732; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CW3: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1050–1050. Prerequisite pointer invoked at proof line(s) 1764; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05PD: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1163–1163. Prerequisite pointer invoked at proof line(s) 1772; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05LW: sources/stacks-chow/tags.txt, source lines 7250–7250. Prerequisite pointer invoked at proof line(s) 1766; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05MH: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 1233–1233. Prerequisite pointer invoked at proof line(s) 1708; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CVV: sources/stacks-chow/tags.txt, source lines 16150–16150. Prerequisite pointer invoked at proof line(s) 1759; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02WY: sources/stacks-chow/tags.txt, source lines 3821–3821. Prerequisite pointer invoked at proof line(s) 1724; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 04AK: sources/stacks-chow/tags.txt, source lines 5619–5619. Prerequisite pointer invoked at proof line(s) 1786; it names a

standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 05MH. Assume*

(1) *f is of finite presentation, and*
(2) *$\mathcal{G}$ is of finite presentation, flat over B, and pure relative to B.*

Then F_{zero} is an algebraic space and $F_{zero} \to B$ is a closed immersion. If $\mathcal{F}$ is of finite type, then $F_{zero} \to B$ is of finite presentation.

Proof. By Lemma 05L6 the module $\mathcal{G}$ is universally pure relative to B. In order to prove that F_{zero} is an algebraic space, it suffices to show that $F_{zero} \to B$ is representable, see Spaces, Lemma 02WY. Let $B' \to B$ be a morphism where B' is a scheme and let $u' : \mathcal{F}' \to \mathcal{G}'$ be the pullback of u to $X' = X_{B'}$. Then the associated functor F'_{zero} equals $F_{zero} \times_B B'$. This reduces us to the case that B is a scheme.

Assume B is a scheme. We will show that F_{zero} is representable by a closed subscheme of B. By Lemma 05MJ and Descent, Lemmas 03I0 and 02W5 the question is local for the étale topology on B. Let $b \in B$. We first replace B by an affine neighbourhood of b. Choose a diagram

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ \downarrow & & \downarrow \\ B & \longleftarrow & B' \end{array}$$

and $b' \in B'$ mapping to $b \in B$ as in Lemma 05KW. As we are working étale locally, we may replace B by B' and assume that we have a diagram

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ & \searrow \quad \swarrow & \\ & B & \end{array}$$

with B and X' affine such that $\Gamma(X', g^*\mathcal{G})$ is a projective $\Gamma(B, \mathcal{O}_B)$-module and $g(|X'|) \supset |X_b|$. Let $U \subset X$ be the open subspace with $|U| = g(|X'|)$. By Divisors on Spaces, Lemma 0CVV the set

$$E = \{t \in B : \mathrm{Ass}_{X_t}(\mathcal{G}_t) \subset |U_t|\} = \{t \in B : \mathrm{Ass}_{X/B}(\mathcal{G}) \cap |X_t| \subset |U_t|\}$$

is constructible in B. By Lemma 0CW3 part (2) we see that E contains $\mathrm{Spec}(\mathcal{O}_{B,b})$. By Morphisms, Lemma 05LW we see that E contains an open neighbourhood of b. Hence after replacing B by a smaller affine neighbourhood of b we may assume that $\mathrm{Ass}_{X/B}(\mathcal{G}) \subset g(|X'|)$.

From Lemma 05PD it follows that $u : \mathcal{F} \to \mathcal{G}$ is injective if and only if $g^*u : g^*\mathcal{F} \to g^*\mathcal{G}$ is injective, and the same remains true after any base change. Hence we have reduced to the case where, in addition to the assumptions in the theorem, $X \to B$ is a morphism of affine schemes and $\Gamma(X, \mathcal{G})$ is a projective $\Gamma(B, \mathcal{O}_B)$-module. This case follows immediately from Lemma 083L.

We still have to show that $F_{zero} \to B$ is of finite presentation if $\mathcal{F}$ is of finite type. This follows from Lemma 083J combined with Limits of Spaces, Proposition 04AK. □